AIDAN BRADSHAW

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EDUCATION

Northwestern University, Evanston, IL — September 2025–Present
Ph.D. in Computer Science
Research in transfer learning, health sensing, mobile and embedded machine learning.
Advisor: Karan Ahuja

Carnegie Mellon University, Pittsburgh, PA — May 2025
M.S. in Applied Data Science | GPA: 4.1/4.0
Research on interpretable medical diffusion in radiology.
Advisor: Adam Perer

San Diego State University, San Diego, CA — May 2024
B.S. in Computer Science (with distinction) | Minor in Applied Mathematics | GPA: 3.81/4.0
Research on mobile health, human-centered computing, and multimodal sensing systems.
Advisor: Shangping Ren

PROFESSIONAL EXPERIENCE

Northwestern University, Chicago, IL — September 2025–Present
Graduate Student Researcher | SPICE Laboratory
Transfer learning methods for wearable activity recognition from IMU and vision data.

Swiss Federal Institute of Technology Zurich (ETH Zurich), Zurich, Switzerland — May 2025–January 2026
Research Assistant | Integrated Systems Laboratory, Digital Circuits and Systems Group
Developed MuyBridge, a mobile computing method for monocular human center-of-mass estimation.

Parsons Corporation, Pittsburgh, PA — January 2025–May 2025
Artificial Intelligence Consultant
Developed policy based data generation and reinforcement learning with human feedback models for internal LLM prompt detection.

Carnegie Mellon University, Pittsburgh, PA — August 2024–May 2025
Research Assistant | School of Computer Science, Human-Computer Interaction Institute
Developed XAI methods for text-to-image diffusion in radiology.

Massachusetts Institute of Technology, Cambridge, MA — May 2024–November 2024
Research Assistant | MIT Media Lab, Responsive Environments Group
Developed low-power on-device bioacoustic classification methods for real-time monitoring of endangered bumblebee species.

AirHop Communications AI, San Diego, CA — June 2023–August 2023
Software Engineer
Developed system-level software for self-organizing network optimization and cell tower outage diagnostics.

San Diego State Research Foundation, San Diego, CA — March 2023–August 2024
Human-Centered Computing Researcher | Computer Architecture and Systems Laboratory
Developed a customizable mobile health platform for Raynaud's monitoring with multimodal symptom, wearable, thermal, and environmental sensing.

AWARDS & HONORS

Murray Fellowship ($48K) | Northwestern University — 2025

Winner, Statistical Machine Learning Final Contest | Carnegie Mellon University — 2025

Best Applied Linear Models Project | Carnegie Mellon University — 2024

Elected to Phi Beta Kappa Honors Society | San Diego State University — 2024

PUBLICATIONS

Peer-Reviewed Conference Papers

7. Yiquan Li*, **Aidan Bradshaw***, Jesse Gao, and Karan Ahuja. 2026. Towards General-Purpose Inertial Foundation Models. Submitted to *NeurIPS 2026*. (*Equal contribution)

6. **Aidan Bradshaw**, Riku Arakawa, Xin Liu, and Karan Ahuja. 2026. TransfHAR: Self-Supervised Wrist Representations for Low-Effort Activity Recognition. Submitted to *ACM UIST 2026*.

5. **Aidan Bradshaw**, Elif Basokur, Marco Giordano, Luca Benini, and Christoph Leitner. 2026. MuyBridge: Quantized 2.5D Network Fusion for On-Device Gait Estimation. Submitted to *ECCV 2026*.

4. Sawyer T. Jones, **Aidan Bradshaw**, Ramaz Tskhadadze, Ben Shen, and Shangping Ren. 2025. A Customizable, Real-Time Mobile Health Application for Raynaud's Syndrome and Beyond. In *Proceedings of IEEE HealthCom 2025*.

3. Katelyn Morrison, Arpit Mathur, **Aidan Bradshaw**, Tom Wartmann, Steven Lundi, Afrooz Zandifar, Weichang Dai, Kayhan Batmanghelich, Motahhare Eslami, and Adam Perer. 2025. A Human-Centered Approach to Identifying Promises, Risks, and Challenges of Text-to-Image Generative AI in Radiology. In *Proceedings of AIES 2025*.

2. **Aidan Bradshaw**, Katelyn Morrison, Arpit Mathur, Weichang Dai, Motahhare Eslami, Kayhan Batmanghelich, and Adam Perer. 2025. Toward Interpretable 3D Diffusion in Radiology: Token-Wise Attribution for Text-to-CT Synthesis. In *Proceedings of MIDL 2025*. Short paper.

1. **Aidan Bradshaw**, Ramaz Tskhadadze, Shangping Ren, and Ben Shen. 2024. A Tailored Health Application: Monitoring the Etiology of Raynaud's Disease. In *Proceedings of CSCSU 2024*.

Peer-Reviewed Journal Papers

1. Patrick Chwalek, Marie Kuronaga, Marco Giordano, **Aidan Bradshaw**, Isamar Zhu, Marina Arbetman, and Joseph A. Paradiso. 2025. Autonomous Low-Power Distributed Acoustic System for Detecting Endangered *Bombus dahlbomii in situ*. Submitted to *Scientific Reports*.

Peer-Reviewed Poster/Demo Papers

1. Katelyn Morrison, Arpit Mathur, **Aidan Bradshaw**, Tom Wartmann, Steven Lundi, Afrooz Zandifar, Weichang Dai, Kayhan Batmanghelich, Motahhare Eslami, and Adam Perer. 2024. Opportunities and Challenges in Designing Text-to-Image Generative AI for Medical Education, Training, and Practice. In *Proceedings of the Pitt AI in Healthcare Research Symposium 2024*.

TEACHING EXPERIENCE

Graduate Teaching Assistant, *Reasoning with Data II*, Carnegie Mellon University 2025

Graduate Teaching Assistant, *Reasoning with Data I*, Carnegie Mellon University 2024

INVITED TALKS

Gest Lecturer, Machine Learning & Sensing Course, Northwestern University 2026

Invited Panelist, Conformable Decoders, Northwestern University 2026

Guest Presenter, Feinberg School of Medicine, Northwestern University 2025

STUDENTS MENTORED

Bhavi Barnwal, Masters, *Computer Science*, Northwestern University 2026

Wenpeng Cai, Undergraduate, *Computer Science*, Northwestern University 2026

Elara Liu, Masters, *Computer Science*, Northwestern University → PhD Student @ Emory 2025

SERVICE

External Reviewer, ACM UIST, ACM IMWUT 2026

Team Captain, SDSU Division I Club Lacrosse 2024

Facilitator and Group Lead, Orange County OCD Support Group, OCD Southern California, Gateway Therapeutic Group, USC OCD Center

Volunteer Work, Someone Cares Soup Kitchen, Aztec Rock Hunger, Breast Cancer Awareness Carnival